King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \tan^{-1}(\sqrt{x})\,dx =$

(a) $\tan^{-1}(x) - x\sqrt{x} + c$

(b) $x\tan^{-1} x - \sqrt{x} + x + c$

(c) $(1 + \tan^{-1}(\sqrt{x}))x + \dfrac{1}{x} + c$

(d) $x\tan^{-1}(\sqrt{x}) - \sqrt{x} + \tan^{-1}(\sqrt{x}) + c$

(e) $\sqrt{x}\tan^{-1}(\sqrt{x}) - \sqrt{x} + c$

2. $\int \frac{x^2+1}{(x-3)(x-2)^2}\,dx =$

(a) $\dfrac{9}{10}\ln\left|\dfrac{x-3}{x-2}\right| + \dfrac{5}{(x-2)^2} + c$

(b) $\dfrac{9}{10}\ln|x-3| + \dfrac{10}{9}\ln|x-2| + \dfrac{5}{(x-2)^2} + c$

(c) $\dfrac{10}{9}\ln|x-2| + 5\ln|x-3| + \dfrac{1}{(x-2)^2} + c$

(d) $10\ln|x-3| - 9\ln|x-2| + \dfrac{5}{x-2} + c$

(e) $9\ln|x-3| + 10\ln|x-2| - \dfrac{5}{x-2} + c$

3. $\int \tan^3 x\,dx =$

(a) $\ln|\sec x + \tan x| + c$

(b) $\dfrac{1}{4}(\tan x)^4 + c$

(c) $\tan x \sec^2 x + c$

(d) $\dfrac{1}{2}\tan^2 x - \ln|\sec x| + c$

(e) $\tan x - \ln|\tan x| + c$

4. The volume of the solid obtained by rotating the region, in the first quadrant, bounded by $y = x$, $y = \dfrac{2x}{1+x^3}$ about $x = -1$, is given by:

(a) $\pi \int_0^2 (x+1)\left(\dfrac{2x}{1+x^3} - x\right) dx$

(b) $\pi \int_{-1}^1 (-x-1)\left(\dfrac{2x}{1+x^3} - x\right) dx$

(c) $2\pi \int_0^1 (x+1)\left(\dfrac{2x}{1+x^3} - x\right) dx$

(d) $2\pi \int_{-1}^1 (x+1)\left(x - \dfrac{2x}{1+x^3}\right) dx$

(e) $2\pi \int_0^1 (x-1)\left(\dfrac{2x}{1+x^3} + x\right) dx$

5. The improper integral $\int_0^\infty \sin^2 x\, dx$

(a) Converges to $\dfrac{1}{3}$

(b) Diverges to $\pm\dfrac{1}{3}$

(c) Diverges to $+\infty$

(d) Converges to $\dfrac{1}{\sqrt{3}}$

(e) Diverges to $-\infty$

6. $\int_0^\infty \frac{e^x}{e^{2x}+3}\, dx =$

(a) $\dfrac{2\pi}{3\sqrt{3}}$

(b) ∞

(c) $3\pi\sqrt{e}$

(d) $\dfrac{\pi}{3\sqrt{3}}$

(e) $\dfrac{3}{2}\pi\sqrt{e}$

7. $\int \frac{\sqrt{1+x^2}}{x}\,dx =$

(a) $\frac{1}{x} - \sqrt{1+x^2} + \frac{1}{2}\ln(1+x^2) + c$

(b) $\ln\left|\frac{1}{x} - \sqrt{1+x^2}\right| + x + c$

(c) $\frac{\sqrt{1+x^2}}{x} + \ln\left|\frac{1+\sqrt{x^2+1}}{x}\right| + c$

(d) $\ln\left|x\sqrt{1+x^2}\right| - \frac{1}{x} + c$

(e) $\ln\left|\frac{\sqrt{1+x^2}-1}{x}\right| + \sqrt{1+x^2} + c$

8. If the fraction $\frac{10}{x^3+9x-x^2-9}$ can be decomposed into two subfractions $\frac{A}{x+K}$ and $\frac{Bx+C}{x^2+9}$, then $K+A+B+C =$

(a) -2

(b) 1

(c) 2

(d) -1

(e) 0

9. $\int x\ln(1+x)\,dx =$

(a) $x^2\ln(x+1) - x\ln(1+x) + c$

(b) $x\ln(1+x) - x^2 - x + c$

(c) $x\ln(1+x) + x^2 + \ln(1+x) + c$

(d) $\frac{1}{2}x^2\ln(x+1) - \frac{1}{2}x^2 + x + c$

(e) $\frac{1}{2}(x^2-1)\ln(1+x) - \frac{1}{4}x^2 + \frac{1}{2}x + c$

10. The volume of the solid obtained by rotating about the x-axis, the region bounded by $x+y=3$, $x=4-(y-1)^2$ is

(a) $\frac{31}{3}\pi$

(b) $\frac{81}{4}\pi$

(c) $\frac{13}{4}\pi$

(d) $\frac{16}{3}\pi$

(e) $\frac{27}{2}\pi$

11. $\int_0^{1/2} x\cos(\pi x)\,dx =$

(a) $\frac{1}{2\pi}-\frac{1}{\pi^2}$

(b) $\frac{\pi-1}{3}$

(c) $\frac{2\pi-1}{3}$

(d) $\frac{1-\pi}{2}$

(e) $\frac{1+\pi}{\pi}$

12. The average value of $f(\theta)=\sec^2\left(\frac{\theta}{2}\right)$ over $\left[0,\frac{\pi}{2}\right]$ is equal to:

(a) $\frac{4}{\pi}$

(b) $\frac{2}{\pi}$

(c) $\frac{\pi}{2}$

(d) 2

(e) 4

13. $\int_0^1 \frac{x^3}{\sqrt{4+x^2}}\,dx =$

(a) 0

(b) $\frac{16}{3} - \frac{7}{3}\sqrt{5}$

(c) $10\sqrt{5} - \frac{1}{3}$

(d) $\frac{8 - \sqrt[3]{7}}{3}$

(e) $5\sqrt{5} - \frac{8}{3}\sqrt{7}$

14. $\int_{\sqrt{2}}^2 \frac{dx}{x^3\sqrt{x^2-1}} =$

(a) $\frac{\pi}{24} + \frac{\sqrt{3}}{8} - \frac{1}{4}$

(b) $\frac{\pi}{13} - \frac{\sqrt{3}}{8} + 1$

(c) $\frac{\pi}{2} - 2\sqrt{2} + \frac{1}{4}$

(d) $2\sqrt{2}\pi + \frac{1}{12}$

(e) 0

15. $\int \sin^6 x \cos^3 x\,dx =$

(a) $-\frac{(\sin x)^7}{7} + \frac{(\sin x)^9}{9} + c$

(b) $\frac{(\sin x)^7}{7} - \frac{(\sin x)^9}{9} + c$

(c) $\frac{(\sin x)^3}{3} + \frac{(\sin x)^6}{6} + c$

(d) $\frac{(\sin x)^5}{5} + \frac{(\sin x)^7}{7} - \frac{(\sin x)^9}{9} + c$

(e) $\frac{(\sin x)^7}{7} - \frac{(\sin x)^9}{9} + \frac{(\sin x)^3}{3} + c$

16. $\int \frac{x^2+x+1}{(x^2+1)^2}\,dx =$

(a) $\tan^{-1}(x^2+x+1) - \frac{1}{2}(x^2+1) + c$

(b) $2\tan^{-1}(x) - \frac{1}{2}\ln(x^2+1) + c$

(c) $\tan^{-1}(x^2+1) - \ln|x| + c$

(d) $\tan^{-1}(x) - \frac{1}{2(x^2+1)} + c$

(e) $\frac{1}{2}\tan^{-1}(x^2+1) - \ln|x^2+x+1| + c$

17. $\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$ converges to:

(a) $\sqrt{2e}$

(b) $2\sqrt{e}$

(c) $\frac{1}{2}e$

(d) $\frac{2}{e}$

(e) $\sqrt{e}$

18. $\int_0^{\pi/4} \sqrt{1+\cos 4x}\,dx =$

(a) $4+\sqrt{2}$

(b) $\frac{\sqrt{2}}{2}$

(c) $2\sqrt{2}$

(d) $2+\sqrt{2}$

(e) $\sqrt{2}$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$t=\sqrt{x}$, by parts: $x\tan^{-1}\sqrt{x}-\sqrt{x}+\tan^{-1}\sqrt{x}+c$.
2	**(d)**	$A=10$, $B=-9$, $C=-5$: $10\ln\lvert x-3\rvert-9\ln\lvert x-2\rvert+\frac{5}{x-2}+c$.
3	**(d)**	$\tan^3 x=\tan x\sec^2 x-\tan x$: $\frac{1}{2}\tan^2 x-\ln\lvert\sec x\rvert+c$.
4	**(c)**	Shells, radius $x+1$, height $\frac{2x}{1+x^3}-x$, $x\in[0,1]$.
5	**(c)**	$\int_0^t \sin^2 x\,dx=\frac{t}{2}-\frac{\sin 2t}{4}\to+\infty$: diverges to $+\infty$.
6	**(d)**	$u=e^x$: $\frac{1}{\sqrt{3}}\left[\tan^{-1}\frac{u}{\sqrt{3}}\right]_1^{\infty}=\frac{\pi}{3\sqrt{3}}$.
7	**(e)**	$x=\tan\theta$: $\ln\lvert\csc\theta-\cot\theta\rvert+\sec\theta$, i.e. $\ln\left\lvert\frac{\sqrt{1+x^2}-1}{x}\right\rvert+\sqrt{1+x^2}+c$.
8	**(a)**	$(x-1)(x^2+9)$: $K=-1$, $A=1$, $B=-1$, $C=-1$, sum $=-2$.
9	**(e)**	By parts then division: $\frac{1}{2}(x^2-1)\ln(1+x)-\frac{1}{4}x^2+\frac{1}{2}x+c$.
10	**(e)**	Shells: $V=2\pi\int_0^3 y(3y-y^2)\,dy=\frac{27}{2}\pi$.
11	**(a)**	By parts: $\left[\frac{x}{\pi}\sin\pi x\right]_0^{1/2}-\frac{1}{\pi^2}=\frac{1}{2\pi}-\frac{1}{\pi^2}$.
12	**(a)**	$\frac{2}{\pi}\left[2\tan\frac{\theta}{2}\right]_0^{\pi/2}=\frac{4}{\pi}$.
13	**(b)**	$u=4+x^2$: $\frac{1}{2}\int_4^5(u^{1/2}-4u^{-1/2})du=\frac{16}{3}-\frac{7}{3}\sqrt{5}$.
14	**(a)**	$x=\sec\theta$: $\int_{\pi/4}^{\pi/3}\cos^2\theta\,d\theta=\frac{\pi}{24}+\frac{\sqrt{3}}{8}-\frac{1}{4}$.
15	**(b)**	$u=\sin x$: $\int(u^6-u^8)du=\frac{\sin^7 x}{7}-\frac{\sin^9 x}{9}+c$.
16	**(d)**	Split: $\frac{1}{x^2+1}+\frac{x}{(x^2+1)^2}$ gives $\tan^{-1}x-\frac{1}{2(x^2+1)}+c$.
17	**(d)**	$u=\sqrt{x}$: $2\int_1^{\infty}e^{-u}du=\frac{2}{e}$.
18	**(b)**	$\sqrt{1+\cos 4x}=\sqrt{2}\cos 2x$: $\sqrt{2}\cdot\frac{1}{2}=\frac{\sqrt{2}}{2}$.